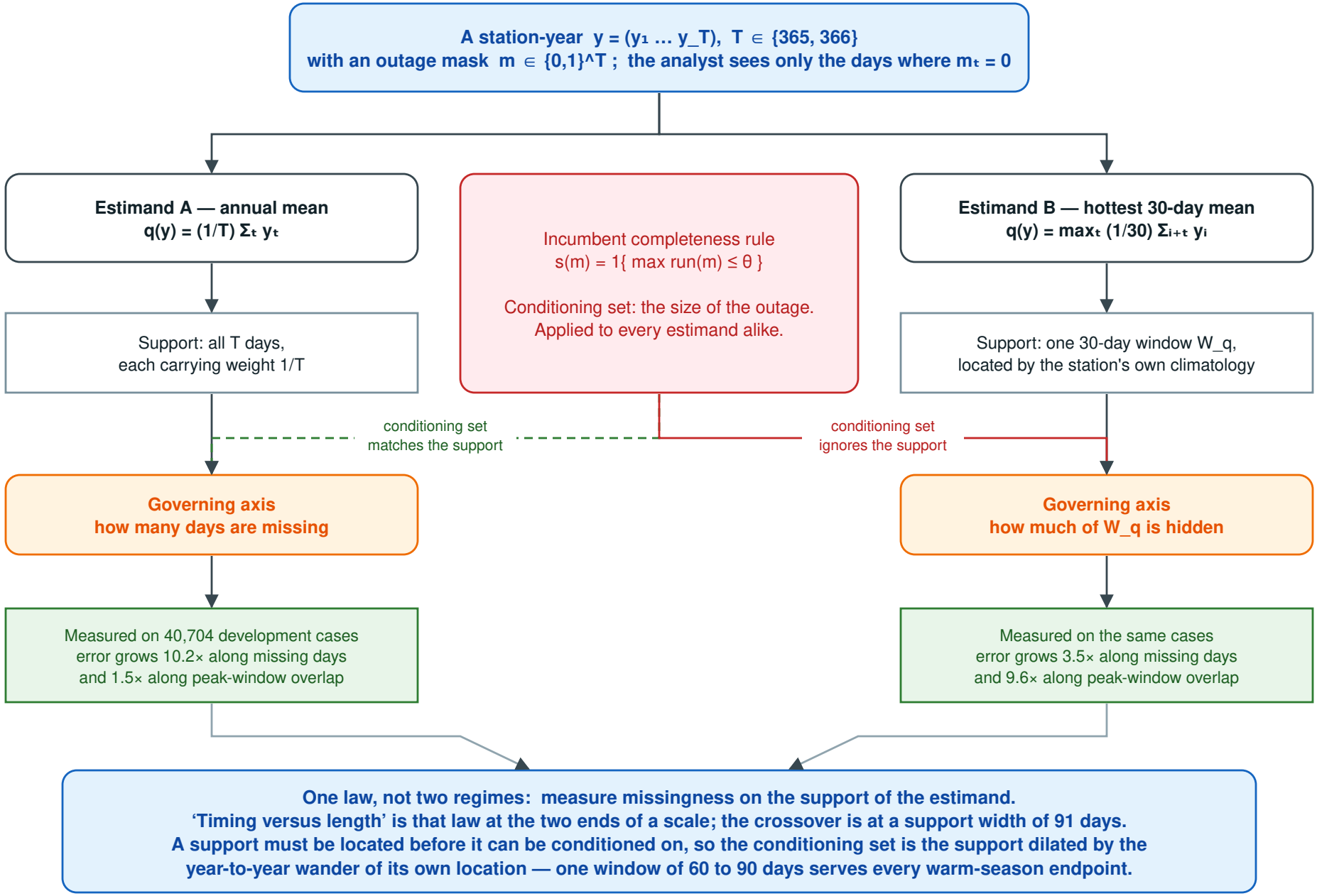
A station-year y = (y₁ … y_T), T ∈ {365, 366}
with an outage mask m ∈ {0,1}^T ; the analyst sees only the days where mₜ = 0
Estimand A — annual mean
q(y) = (1/T) Σₜ yₜ
Incumbent completeness rule
s(m) = 1{ max run(m) ≤ θ }
Conditioning set: the size of the outage.
Applied to every estimand alike.
Estimand B — hottest 30-day mean
q(y) = maxₜ (1/30) Σᵢ₊ₜ yᵢ
Support: all T days,
each carrying weight 1/T
Support: one 30-day window W_q,
located by the station's own climatology
conditioning set
matches the support
conditioning set
ignores the support
Governing axis
how many days are missing
Governing axis
how much of W_q is hidden
Measured on 40,704 development cases
error grows 10.2× along missing days
and 1.5× along peak-window overlap
Measured on the same cases
error grows 3.5× along missing days
and 9.6× along peak-window overlap
One law, not two regimes: measure missingness on the support of the estimand.
'Timing versus length' is that law at the two ends of a scale; the crossover is at a support width of 91 days.
A support must be located before it can be conditioned on, so the conditioning set is the support dilated by the
year-to-year wander of its own location — one window of 60 to 90 days serves every warm-season endpoint.